MILITARY SPECIFICATION

MANROPE, COMPOSITION, COVERED

1. SCOPE

1.1 This specification covers composition manrope, which includes a wire rope core, a plastic filler and a nylon cover, suitable for use as a hand rail for inclined ladders and guard rails not exposed to the weather.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification.

SPECIFICATIONS

FEDERAL
QQ-S-781 - Strapping, Flat; Steel.
RR-W-410 - Wire Rope and Wire Strand.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.

MILITARY
JAN-P-100 - Packaging and Packing for Overseas Shipment General Specification.
MIL-C-132 - Crates, Open, Wood; Maximum Capacity 2,500 Pounds.
MIL-A-140 - Adhesive, Water-Resistant, Waterproof Barrier-Material.

NAVY DEPARTMENT
General Specifications for Inspection of Material.

STANDARDS

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications. - The following document forms a part of this specification. Unless otherwise indicated the issue in effect on date of invitation for bids shall apply.

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, N. Y.)

3. REQUIREMENTS

3.1 Definition of terms. - The following terms for composition manrope is used in this specification:

3.1.1 Core. - The core is the inner or foundation member.

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3.1.2 Filler. - The filler is the intermediate member used to give the manrope its mass or size.

3.2 Materials. - Materials shall be as specified herein. Materials not definitely specified shall be of the best quality used for the purpose in commercial practice. Material shall be free from all defects and imperfections that may affect the serviceability of the manrope.

3.2.1 Core. - The material for the core shall be type I, class 2, 6 by 19 galvanized wire rope in accordance with Specification RR-W-410.

3.2.2 Filler and jacketing. - The filler material shall be of polyvinyl chloride without the addition of compounding ingredients such as pigment and dyes. The material for the jacketing shall be of nylon and white in color. The polyvinyl chloride filler and nylon jacketing shall be of synthetic resin type and shall conform to the requirements shown in table I.

Table I - Material requirements.

Composition	Specific gravity	
	Minimum	Maximum
Polyvinyl chloride	1.2	1.5
Nylon	1.07	1.09

3.3 Design and construction. -

3.3.1 General. - The polyvinyl chloride filler shall be concentrically extruded over a wire rope core and the surface shall be indented or circular grooved, so as to make a mechanical lock between the filler and the nylon jacket. The nylon jacketing shall be concentrically extruded over the filler to insure a uniform cross-section. The wire rope core shall be under tension of not less than 600 pounds when the filler material is applied.

3.3.2 Size. - The manrope shall be manufactured in one size only with an overall diameter of 7/8 inch, plus or minus 1/32 inch. The core shall have a diameter of 1/4 inch. The filler shall have an outside diameter of 3/4 inch and covered with 1/16 inch thick nylon as specified in 3.3 1, and shall be designed to withstand the tests specified in 4.5.

3.3.3 Solvent and weather resistance. - Perspiration, oils and detergents shall have no ill effect on either plastic layer when tested as specified in 4.5.1.

3.4 Defects. - When material, which has been examined, tested and accepted, contains inherent defects subsequently exposed, the contractor may be required to replace the defective material without expense to the Government.

3.5 Workmanship. - The manrope shall be manufactured in accordance with the best commercial practice and consistent with the best rigging practice. The workmanship shall be free from any defects which may effect the serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified herein the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

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4.2 Sampling for lot acceptance. -

4.2.1 Lot. - Unless otherwise specified in the contract or order, a lot shall consist of all manrope presented for delivery at one time and on the same contract or order.

4.2.2 Sampling for visual and dimensional examination. - A random sample length of manrope shall be selected from each lot offered for examination in accordance with level III of Standard MIL-STD-105.

4.2.3 Sampling for tests. -

4.2.3.1 When the manrope can be identified by its manufacturing reel, coil or bundle, one sample 18 inches long shall be taken from a shipping reel, coil or bundle in the lot cut from the same manufacturing reel and submitted for the tests specified in 4.4.

4.2.3.2 When the manrope cannot be identified by its manufacturing reel, coil or bundle, one sample 18 inches long shall be taken from each reel, coil or bundle submitted for tests specified in 4.4.

4.3 Visual and dimensional examination. - Each of the sample lengths selected in accordance with 4.2.2 shall be visually and dimensionaly examined to verify compliance with the requirements of this specification not involving tests. Any length in the sample containing one or more visual or dimensional defects shall be rejected and if the number of defective lengths in any sample exceed the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.4 Tests. - Tests shall be performed at the point of manufacture under the supervision of the Government inspector. They shall consist of tests specified in 4.5.

4.4.1 Rejection. - Any reel, coil or bundle of manrope found to be nonconforming shall be rejected.

4.5 Test methods. -

4.5.1 Weathering. - Test for weathering shall be performed by use of an Atlas Twin Arc Weather-O-Meter and shall be exposed for 100 hours without ill effect.

4.5.2 Hardness. - Hardness shall be conducted at 270° F. for nylon and 200° F. for polyvinyl chloride. These tests are to be made with the Fisher M. P. apparatus. Also, the test specimens shall be subjected to a -20° F. temperature for at least 4-hours, then bent over a 4-inch diameter mandrel, for 90 degrees and then back 180 degrees (at no specified time) without impairing the material for general use.

4.6 Retest. - In case of failure of samples of the manrope, retests may be permitted at the discretion of the Government inspector.

4.7 Inspection procedures. - For Naval purchases, the general inspection procedures shall be in accordance with the General Specifications for Inspection of Material.

5. PREPARATION FOR DELIVERY

5.1 Packaging. -

5.1.1 Level A. - Manrope shall be furnished in reels, coils or bundles as specified (see 6.1) as follows:

5.1.1.1 Reels. - Manrope in length as specified (see 6.1) and wound on wooden reels shall be protected by waterproof paper that extends completely around the reel. The waterproof paper wrap shall be tacked to the flanges of the reel. Reels shall be completely inclosed with wooden lagging with lagging boards touching each other. Lagging boards are to be the same thickness as the flange, but not thicker than 2-inch commercial lumber. Boards shall be nailed to the outside of flanges of the reel extending across each flange and secured with a minimum of two flat straps conforming to type I, class B of Specification QQ-S-781, drawn tight around the circumference of the reel and stapled.

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5.1.1.2 Coils.- Manrope in length as specified (see 6.1) furnished in coil form shall be wrapped with waterproof paper and securely tied with a minimum of four ties equally spaced. Fiber cushion protectors shall be placed under ties to protect the wrap and contents.

5.1.1.3 Bundles.- Manrope in cut lengths as specified (see 6.1) and furnished in bundles shall be individually wrapped with waterproof paper. Closure of the wrap, seams and ends, shall be effected by use of adhesive conforming to Specification MIL-A-140. Wrap shall be tied with a minimum of four ties equally spaced. Fiber cushion protectors shall be placed under ties to protect the wrap and contents.

5.1.2 Level C.- Manrope in lengths as specified (see 6.1) shall be packaged in accordance with the manufacturers commercial practice.

5.2 Packing.-

5.2.1 Level A.-

5.2.1.1 Coils and bundles.- Coils and bundles of like lengths shall be packed in nailed wood or wood-cleated plywood boxes or sheathed crates conforming to Specification PPP-B-621 (class 2), PPP-B-601 (overseas type) or MIL-C-104, respectively at the option of the contractor. Wood or wood-cleated boxes packed with contents exceeding gross weight of 250 pounds shall be modified by the addition of skids in accordance with the applicable box specification.

5.2.1.2 Reels.- Reels will require no additional packing. When specified (see 6.1) reels shall be packed in unsheathed crates conforming to Specification MIL-C-132.

5.2.2 Level B.-

5.2.2.1 Coils and bundles.- Coils and bundles of like lengths shall be packed in nailed wood or wood-cleated plywood boxes or sheathed crates conforming to Specification PPP-B-621 (class 1), PPP-B-601 (domestic type) or MIL-C-104, respectively, at the option of the contractor. Wood or wood cleated boxes packed with contents exceeding gross weight of 250 pounds shall be modified by the addition of skids in accordance with the applicable box specification.

5.2.2.2 Reels.- Reels will require no additional packing.

5.2.3 Level C.- Manrope of like lengths shall be packed in a manner to insure acceptance by common carrier and safe delivery to destination. Containers shall conform to the Uniform Freight Classification Rules or other applicable regulations as applicable to the mode of transportation at the lowest rate to the point of delivery.

5.3 Blocking, bracing and cushioning.- Blocking, bracing and cushioning shall be in accordance with Specification JAN-P-100 or MIL-C-104 and appendix thereto.

5.4 Marking.- In addition to any special marking required by the contract or order, reels, coil, bundles and shipping containers shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Ordering data.- Procurement document should specify the following:

(a) Title, number, and date of this specification.
(b) Length required.
(c) Whether packaging shall be level A or C (see 5.1.1 and 5.1.2); Whether packaging shall be level A, B or C (see 5.2.1, 5.2.2 and 5.2.3).
(d) Whether crates are required (see 5.2.1.2).

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6.2 Cover.- The cover is the protective nylon jacketing extruded over the polyvinyl chloride filler.

Notice.- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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